SkillNavora

Audit

Course gap audit

The master prompt checked topic by topic against the existing course, with the status of every area.

Course gap check: supplement to the Data Science, AI and Data Engineering course. Existing lessons are unchanged.

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Course gap audit

This audit checks the Data Science, AI and Data Engineering master prompt against the existing course, topic by topic. Each row says whether the area is covered, partly covered or missing, and what the supplement adds.

Summary: 20 areas covered, 9 partly covered, 2 missing. The three supplements fill the missing and partial areas. Existing lessons, PDFs and tabs were not changed.

Coverage by prompt module

Prompt module	Course home	Status	What the course covers	What is missing or partial	Where the supplement covers it
1. Advanced Excel for Data Analysis	Course Module 1: Advanced Excel (16 lessons)	COVERED	Fundamentals, formatting, validation, formulas, text/logical/IF, sorting and dynamic arrays, XLOOKUP/VLOOKUP/INDEX-MATCH, pivot tables, dates and named ranges, charts, Superstore dashboard, Power Query, Power Pivot, What-If. Added in lessons 13-16: Flash Fill, Text to Columns and data standardising; macros and VBA; Office Scripts, Power Automate and AI formula testing; business projects (complaints, finance tracker, productivity, customer segmentation).	None from Supplement B1.	Supplement B, section B1
2. SQL for Data Analysis and Database Management	Course Module 2: SQL (20 lessons)	PARTIAL	Databases and RDBMS, keys and normalisation, DDL/DML, SELECT and filters, functions, GROUP BY/HAVING/CASE, joins, set operators, subqueries, window functions, CTEs, views, procedures, transactions, indexes, end-to-end case study. Added in lessons 16-20: recursive CTEs and hierarchies; percentiles and distributions; business SQL (cohorts, funnels, churn, CLV, retention); incremental loading and data-quality SQL; SQL across SQL Server, BigQuery, Snowflake and Databricks SQL.	Missing: partitioning, and OLTP vs OLAP in depth (lesson 20 covers the platforms only).	Supplement B, section B2

3. Power BI and Business Intelligence	Course Module 3: Power BI (8 lessons)	**PARTIAL**	Power BI start, Power Query and star schema, DAX fundamentals, DAX time intelligence, visuals and drill-through, service, connectivity modes, security and a capstone. Added in lessons 7-8: Microsoft Fabric (OneLake, lakehouse, warehouse, dataflows, semantic models) and Databricks (notebooks, Delta Lake, Spark jobs, Unity Catalog).	Missing: Copilot inside Power BI as a lesson, deployment pipelines in depth, and incremental refresh. Industry dashboards (HR, finance, marketing, operations) are only examples.	Supplement A, section A10 (Fabric and platforms); Supplement C (dashboard interview)
4. AI and Data Analysis	Course Module 5: AI for Data Analysis (4 lessons)	**PARTIAL**	GenAI for analysts, prompting, AI with SQL and Text-to-SQL, AI in Excel and Office, the AI-assisted analyst workflow with agents and governance.	Missing: AI anomaly detection and forecasting as hands-on lessons, AI-generated DAX, and a structured tool comparison for Julius AI, NotebookLM and Perplexity.	Supplement C, section C6 (AI analytics checklist)
5. Python Programming for Data Science	Course Module 6: Python for Data Science (31 lessons)	**COVERED**	Python basics to advanced, OOP, files and JSON, environments and Git, testing and debugging, NumPy, pandas I to IV, visualisation, EDA, SQL from Python, APIs and scraping, performance, DSA, retail capstone. Added in lessons 30 and 31: automation with scheduled scripts, logging and email; production Python with type hints, logging and packaging.	None from Supplement B6.	Supplement B, section B6 (production Python)
6. Statistics and Probability	Course Module 7: Statistics & ML (first lessons)	**COVERED**	Descriptive statistics, probability and distributions, sampling and CLT, hypothesis testing (z, t, chi-square, ANOVA, non-parametric), correlation, A/B testing and experiments.	Partly missing: business statistics for revenue forecasting and risk analysis as worked cases.	Covered by the existing lessons; Supplement C adds interview questions
7. Machine Learning and Predictive Analytics	Course Module 7: Statistics & ML (30 lessons)	**COVERED**	Introduction to ML, scikit-learn workflow, preprocessing and features, regression, gradient descent, regularisation, logistic regression, metrics, cross-validation and tuning, KNN, trees, ensembles, SVM and Naive Bayes, imbalanced data, clustering and PCA, time series, NLP, neural networks, XAI and MLOps, churn capstone. Added in lessons 29 and 30: recommendation systems; regression metrics MAE, MSE, RMSE and R-squared with baselines.	None from Supplement B7.	Supplement B, section B7 (recommendation systems)

8. Data Engineering Fundamentals	Data Engineering & Cloud track (15 lessons)	**COVERED**	Only small touches: Parquet and DuckDB in the Python performance lesson, and API ingestion in the APIs lesson. Data engineering roles and file formats (CSV, JSON, XML, Parquet, Avro, ORC); ingestion (batch, streaming, API, CDC); ETL and ELT; idempotent, retried pipelines; data modelling and SCD types 1 to 3.	Hands-on labs are not included: the Airflow, Spark and Kafka code is shown but not executed.	Supplement A, sections A1 and A4; Supplement B9, B10
9. Data Warehousing, ETL and Data Pipelines	Data Engineering & Cloud track (lessons 5 to 9)	**COVERED**	Star schema appears in the Power BI lesson only. Warehouse, data mart, data lake and lakehouse; medallion layers with a quality gate; Airflow DAGs with retries; dbt models, tests and lineage; Spark and PySpark; Kafka topics, consumers and delivery guarantees.	Hands-on labs are not included: Airflow, dbt, Spark and Kafka code is shown but not executed.	Supplement A, sections A2 to A8
10. Cloud Computing and Modern Data Platforms	Data Engineering track (lessons 11, 12); Power BI lessons 7 and 8; SQL lesson 20	**PARTIAL**	Docker is mentioned in the AI SQL lesson; Metabase is shown there too. Added: Microsoft Fabric and Databricks platform concepts (Power BI lessons 7-8) and the SQL platform comparison (SQL lesson 20). Cloud service models, regions, identity, service mapping across AWS, Azure and Google Cloud, and cost control (lesson 11); Docker, Kubernetes and CI/CD (lesson 12); Fabric and Databricks concepts (Power BI lessons 7 and 8); SQL platform comparison (SQL lesson 20).	No hands-on cloud labs; cloud services are mapped and costed but not deployed.	Supplement A, sections A9 to A11
11. Generative AI and Large Language Models	Course GenAI track (12 lessons)	**COVERED**	LLM basics, prompt engineering, calling LLM APIs, embeddings and vector search, RAG, tool use, agents, MCP, evaluation and guardrails, customising models, GenAI for data work, shipping a GenAI app.	Partly missing: context engineering as a lesson of its own, and a provider comparison table for OpenAI, Claude and Gemini with current model names and pricing (to be checked on the day of publishing).	Supplement C, section C6
12. AI Agents and Agentic Workflows	Agents track (lessons 7 to 11)	**COVERED**	Agents, tool calling, MCP, no-code agents, Claude Code, voice agents, multi-agent systems in the GenAI lessons. Added in lessons 10 and 11: agent frameworks LangGraph, CrewAI and AutoGen; agent evaluation and monitoring.	None from Supplement C5.	Supplement C, section C5

13. MCP, APIs and Integrations for PMs and Business Stakeholders	Agents track (lessons 7 to 9)	**COVERED**	APIs without code, REST, JSON, authentication, webhooks, Postman, MCP without code, n8n, Make, Zapier, Replit. Added in lessons 7 to 9: OpenAPI and Swagger; OAuth 2.0 with PKCE and secure tokens; MCP security and business integrations for CRM, ERP, Jira, Slack and Notion.	None from Supplement B8.	Supplement B, section B8
14. AI and No-Code Tools Laboratory	Tools track (3 lessons)	**PARTIAL**	Figma and n8n tutorials; the Claude, n8n, Make and Zapier lesson. Added in lesson 3: the tools and learning resources directory with its sixteen categories, and the tool evaluation sheet.	Full hands-on tutorials for each category are still to be written.	Supplement C, section C7 (tool evaluation sheet)
15. Data Governance, Security and Responsible AI	Governance track (lessons 3 to 5); Data Engineering lesson 10	**COVERED**	Guardrails, prompt injection, fairness and explainability, row-level security, transactions and security basics. Added: data ownership and stewardship; retention rules; GDPR and India's DPDP Act, 2023 principles; masking, pseudonymisation and row-level security; audit; responsible AI checks; threat modelling and incident response; data quality and classification.	The legal detail is for orientation only and is not legal advice; check current rules with counsel.	Supplement B, sections B3 to B5
16. Business Problem-Solving and Analytics Consulting	Governance track (lessons 1 and 2)	**COVERED**	Business framing in the ML introduction; BRD and FRD guides on the site; PM track discovery material. Added: KPIs and driver metrics; process mapping with bottlenecks; stakeholder discovery with Five Whys and a fishbone diagram; hypotheses and recommendations; user stories, acceptance criteria, PRDs, data storytelling and agile delivery.	No live client case studies; the scenarios are illustrative.	Supplement B, sections B3 and B4
17. Capstone Projects and Portfolio Development	Career track (lesson 8); Agents and Data Engineering tracks	**COVERED**	One capstone per track (Excel, SQL, Power BI, Tableau, AI, Python, ML, GenAI, agents), career portfolio and GitHub lessons. Added in career lesson 8: eight capstone briefs with requirements, deliverables and rubrics, and a portfolio checklist.	None from Supplement C4.	Supplement C, section C4 (8 capstone briefs)

18. Interview Preparation and Career Readiness	Career track (lessons 5 to 7, 9)	**COVERED**	Data analyst and data scientist interview preparation, portfolio and GitHub, 90-day plan. Added: data and analytics interview questions with model answers (lesson 5); business scenario interviews (lesson 6); STAR behavioural answers (lesson 7); readiness check (lesson 9).	None from Supplements C1 to C3 and C9.	Supplement C, sections C1 to C3
Part C: 20 cross-functional capabilities	Spread across the course	**PARTIAL**	Git and GitHub, testing and debugging, REST APIs and webhooks, DSA, Python production topics, MLOps (in the ML lessons), LLM evaluation.	Missing: Linux and command line (only basic terminal use in Python lessons), system design fundamentals, orchestration (Airflow), dbt, data observability, cloud cost optimisation, data and AI architecture, technical documentation, Agile and project delivery, production troubleshooting, business communication.	Supplement A sections A4 and A12; Supplement B sections B5, B9 and B10
Part D: 12-month roadmap	Career track (lesson 10)	**COVERED**	A 90-day plan in the career lessons. Added in career lesson 10: a twelve-month beginner roadmap with weekly hours, milestones and assessments.	None from Supplement C8.	Supplement C, section C8
Part E: Tools & Learning Resources directory	Tools track (lesson 3)	**COVERED**	The site has a tools and learning resources page with categorised links. Added in tools lesson 3: the sixteen-category directory structure and the tool evaluation sheet.	None from Supplement C7 (structure).	Supplement C, section C7 (categories and checked dates)
Part G: PDF 1 Master Handbook	Course PDFs: Complete Data Science & AI Handbook Parts 1 to 4	**PARTIAL**	Four complete handbook parts exist.	Needs a check that the parts include every module above (Data Engineering and Governance are not in them).	Supplement A, Supplement B and this audit complete the set
Part G: PDFs 2 to 6 (Excel, SQL, Power BI, Python, Stats & ML)	Course lesson PDFs by track	**COVERED**	One PDF per lesson in each track (Excel 12, SQL 15, Power BI 6, Python 29, Stats & ML 28).	Track-level handbooks combining the lessons are not separate PDFs.	Existing lesson PDFs are unchanged

Part G: PDF 7 Data Engineering and Cloud Handbook	Data Engineering & Cloud track (15 lessons)	**COVERED**	Added as lessons with PDFs (data-engineering lessons 1 to 15).	Hands-on labs as noted for Airflow, Spark and Kafka.	Supplement A (full handbook)
Part G: PDF 8 to 9 (GenAI, RAG, Agents and MCP)	Course GenAI and Agents lesson PDFs	**COVERED**	Lesson PDFs for GenAI (12) and Agents (6).	Track-level handbooks are not separate PDFs.	Existing lesson PDFs are unchanged
Part G: PDF 10 AI and No-Code Tools Laboratory	Tools track (2 lessons)	**PARTIAL**	Figma and n8n.	The tool directory and setup guides for the other tools.	Supplement C, section C7
Part G: PDF 11 Capstone Projects and Portfolio Workbook	Career track (lesson 8)	**COVERED**	Capstone lessons and portfolio lessons. Added: eight capstone briefs and the portfolio checklist (career lesson 8, with PDF).	Workbook templates are not separately provided.	Supplement C, section C4
Part G: PDF 12 Interview Preparation Handbook	Interview-Handbook.pdf and career interview lessons	**COVERED**	Interview handbook and two career interview lessons.	Data engineering and scenario sections.	Supplement C, sections C1 to C3
Part G: PDF 13 12-Month Roadmap and Tracker	Career track (lesson 10)	**PARTIAL**	Added: the roadmap (career lesson 10).	A printable tracker is not yet provided.	Supplement C, section C8
Part G: PDF 14 Quick Reference and Cheat Sheets	NOT IN THE COURSE	**MISSING**	None as one collection.	Excel, SQL, DAX, Python, pandas, statistics, ML metrics, API and MCP terms, AI terms.	Supplement B, section B11 (cheat sheet index)
Part G: PDF 15 Data Science and AI Glossary	NOT IN THE COURSE	**MISSING**	None as one glossary.	Terms and acronyms with simple definitions and examples.	Supplement B, section B11 (glossary index)

Proposed additions for your approval

These are added as supplements, not as edits to existing lessons. You approve each one before any change to the course itself.

- Done: a Data Engineering & Cloud track (15 lessons) with PDFs. Still to add: hands-on labs for Airflow, dbt, Spark, Kafka and one cloud.
- Done: a Governance, Privacy & Business Consulting track (5 lessons) covering B3, B4 and B5.
- Done: Excel automation (B1) as lessons 13 to 16; SQL topics (B2) as lessons 16 to 20; Fabric and Databricks (A10) as Power BI lessons 7 and 8; production Python (B6) as Python lessons 30 and 31; recommenders and regression metrics (B7) as ML lessons 29 and 30; agent APIs, OAuth, MCP and frameworks as Agents lessons 7 to 11.

- Done: the eight capstone briefs (C4) as career lesson 8 and the 12-month roadmap (C8) as career lesson 10. A printable workbook and tracker are still to be added.
- Partly done: the tools directory and evaluation sheet are in the tools track (lesson 3). Hands-on no-code tutorials for each category are still to be written.